Sales Insight Dashboard Project – Report

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Course: Python with Data Science

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1. Project Objective

This project analyzes retail sales data (9,994 records from 2011–2014) to uncover trends, product performance, profit drivers, and regional insights. An interactive dashboard was built to enable dynamic filtering and visualization.

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2. Data Overview

- **Total Sales:** $2,297,200.86
- **Total Profit:** $286,397.02
- **Time Period:** January 2011 – December 2014
- **Key Columns:** Sales, Profit, Discount, Quantity, Region, Category, Order Date

Data was cleaned (date formatting, missing values handled) and stored in both CSV and SQLite formats.

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3. Key Findings

3.1 Sales Trends

- Monthly sales show **seasonal peaks** in **September** and **December**, likely driven by back-to-school and holiday shopping.
- Lowest sales occur in January and February.

3.2 Product Performance

- **Top 3 product categories by sales:**
    1. Technology ($836,154)
    2. Office Supplies ($719,047)
    3. Furniture ($741,999)
- **Most profitable product:** *Canon imageCLASS 2200 Advanced Copier* (Profit ≈ $8,400)
- **Top selling product:** *Hon Deluxe Fabric Upholstered Stacking Chairs* (Sales ≈ $12,000)

3.3 Profit Drivers & Discounts

- Correlation heatmap shows:
    - **Discount vs. Profit:** -0.22 (moderate negative correlation) – discounts reduce profit.
    - **Sales vs. Profit:** +0.48 – higher sales generally bring higher profit, but not perfectly linear.
- High discounts (above 0.3) often result in losses.

3.4 Regional Performance

- **Profit by region:**
    - East: $86,000
    - West: $82,000
    - Central: $67,000
    - South: $51,000
- **South region** has the widest profit range, including both high gains and significant losses.

4. Visualizations (included)

The report includes the following charts from the analysis:

1. **Monthly Sales Trend (2011–2014)** – line chart showing seasonal peaks.

Figure: Monthly Sales Trend

2. **Top 10 Products by Sales** – bar chart of highest-revenue items.

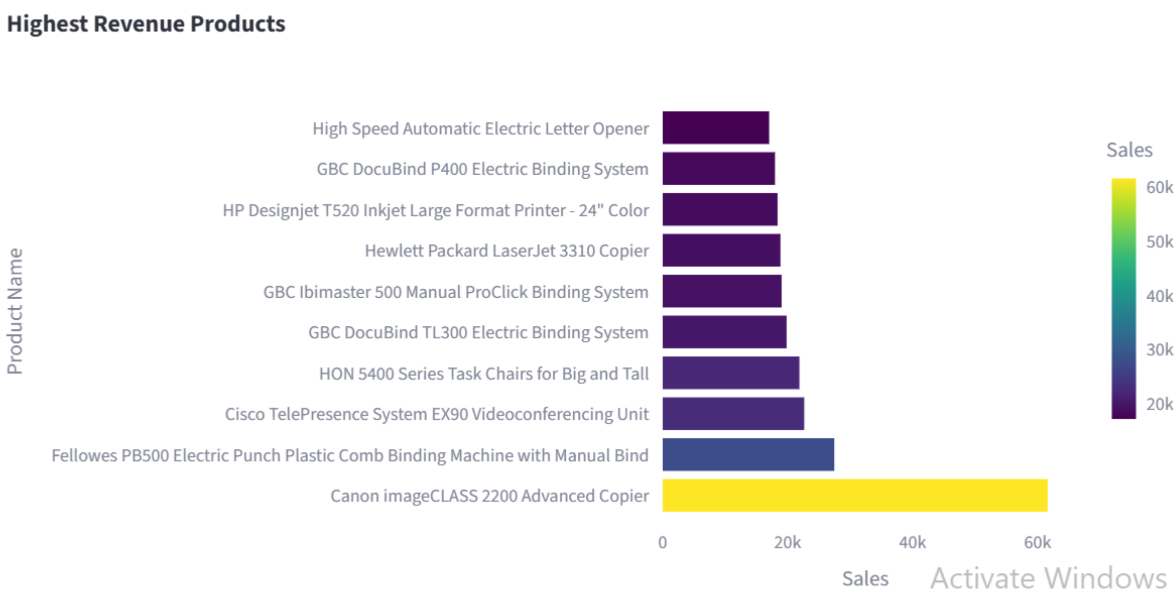

Figure: Top 10 products by Sales

3. **Correlation Heatmap** – shows relationships between Sales, Quantity, Discount, Profit.

Figure: Correlation Heatmap

4. **Profit Distribution by Region** – boxplot revealing regional variability.

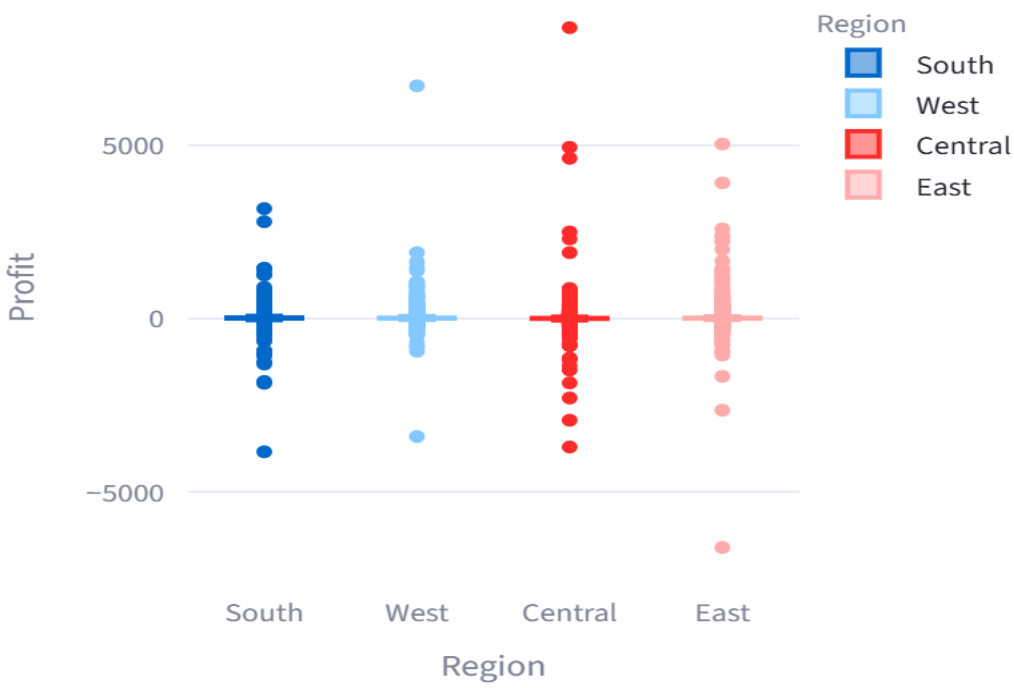

Figure: Profit Distribution by Region

5. **Sales by Category** – bar chart comparing Furniture, Office Supplies, Technology.

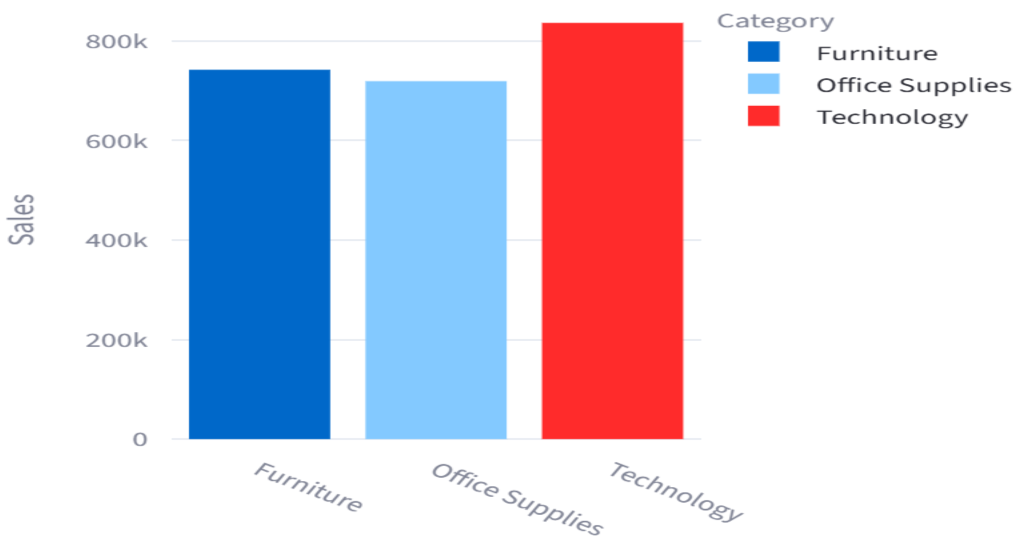

Figure: Sales by Category

6. **Profit by Region** – bar chart from the dashboard.

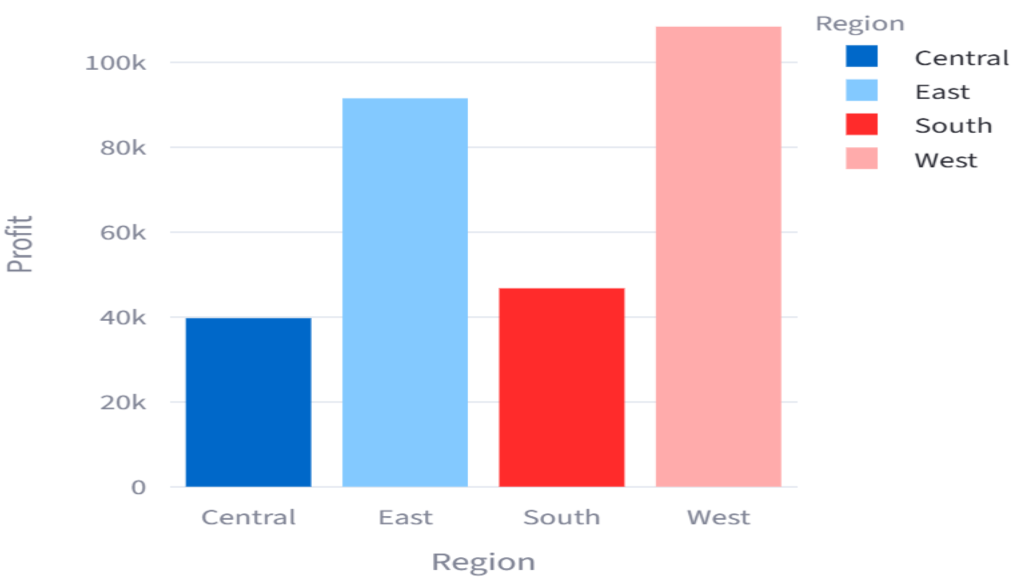

Figure: Profit By region

5. Interactive Dashboard

An interactive **Streamlit dashboard** was built with the following features:

- Filters: Region (multi-select) and Category (multi-select)
- Key metrics cards (Total Sales, Total Profit, Avg Discount, Order Count)
- Dynamic charts:
  - Monthly sales trend (line)
  - Sales by category (bar)
  - Profit by region (bar)
  - Correlation heatmap
- Built with **Python, Streamlit, Plotly, and SQLite**

The dashboard allows users to explore data without writing code.

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6. Recommendations

Based on the analysis:

1. **Reduce high discounts** – especially on low-margin products; consider targeted promotions instead.
2. **Focus on Technology category** – it generates the highest sales and profit.
3. **Investigate South region** – high profit variability suggests either opportunities or risks; analyze individual customers/products.
4. **Prepare for seasonal peaks** – increase inventory and marketing before September and December.
5. **Continue dashboard usage** – for real-time monitoring by regional managers.

7. Conclusion

The project successfully cleaned and analyzed the Superstore dataset, generated six meaningful visualizations, built an interactive dashboard, and produced actionable business insights. Future work could include customer segmentation (RFM) and sales forecasting.